Anurag Nepal

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PROFESSIONAL SUMMARY

Machine learning engineer and data scientist with M.S. in Statistical Data Science (4.0 GPA, Distinguished Graduate). 1.5 years building ML models for cancer genomics and medical imaging, including a peer-reviewed CNN publication. Strong in Python, PyTorch, statistical modeling, and reproducible research. Open to ML engineering and data science roles.

TECHNICAL SKILLS

Languages: Python (PyTorch, TensorFlow, scikit-learn, Keras, OpenCV), R, SQL, Java, C/C++.

Methods: penalized regression (lasso, group lasso), survival analysis, hypothesis testing, false discovery control, CNNs, classification, regression, forecasting, cross-validation, hyperparameter tuning.

Tools: Git, Jupyter, BigQuery, reproducible pipelines, experiment tracking, data visualization.

EXPERIENCE

Graduate Assistant, Dept. of Mathematics, San Francisco State University Feb 2025 - May 2026

- Built statistical ML models for cancer genomics: permutation-assisted group lasso for multi-omics feature selection on TCGA-BRCA data, holding false discovery rate at 0.01 across 18 simulation scenarios (200-400 noise features) vs. 0.53 for standard lasso
- Developed survival-analysis models on breast cancer genomics and clinical data reaching 0.948 AUC; validated with resampling-based evaluation and documented experiments for reproducibility
- Presented statistical genomics findings at BBSW 2025; reproduced Random Forests, Boruta, and variable-importance methods from peer-reviewed papers as validated R code

SELECTED PROJECTS

Thesis Findings: Multi-modal Feature Selection Live Demo

- M.S. thesis on permutation-assisted group lasso for multi-modal feature selection, validated on TCGA breast cancer data

Biomedical Imaging Analysis Live Demo

- COVID-19 lung infection segmentation and quantification on chest CT, plus an in-browser demo

Malaria Detection using Machine Learning Live Demo

- CNN classifier for malaria parasites in blood smear images, plus an in-browser demo (peer-reviewed publication)

Kalshi Backtest Lab Live Demo

- Market data pipeline and statistically rigorous backtesting of Kalshi BTC prediction-market strategies

EDUCATION

San Francisco State University 08/2024 - 05/2026

M.S. in Statistical Data Science, GPA: 4.0/4.0, Distinguished Graduate Award

Nitte Meenakshi Institute of Technology 08/2019 - 06/2023

B.E. in Computer Science and Engineering, GPA: 9.34/10.0, ranked among the top 5 graduates

CERTIFICATIONS

- Data Analyst, Datacamp
- Intermediate SQL, Datacamp
- Python, Coursera

PUBLICATION

- Co-author, "Detection of Malaria in Blood Cells using CNN" - peer-reviewed paper on CNN-based malaria detection from blood-smear images